

# Daniel Foreman-Mackey

Research Scientist, Center for Computational Astrophysics, Flatiron Institute

[dfm@dfm.io](mailto:dfm@dfm.io) | [dfm.io](https://dfm.io) | [github.com/dfm](https://github.com/dfm) | [linkedin.com/in/dan-foreman-mackey](https://linkedin.com/in/dan-foreman-mackey)

Lead developer and core maintainer of many popular open source packages including [emcee](#), [tinygp](#), [corner.py](#), and [showyourwork](#); a full list can be found on [GitHub](#).

As of 2026-08-22<sup>1</sup>, published 4 refereed papers with 63 total citations, and an h-index of 4; a complete listing can be found on [Google Scholar](#).

## Experience

---

### Center for Computational Astrophysics, Flatiron Institute, New York, NY

Sep 2022 – present *Research Scientist*

Sep 2017 – Sep 2022 *Associate Research Scientist*

Associate professor equivalent faculty member. Co-leader of “Astronomical Data” group with a steady state of about 10 full time researchers; managed and mentored 8 postdoctoral research fellows, and 9 graduate student researchers. Developed, released, and maintain [exoplanet](#), a suite of Python libraries for probabilistic inference with large astronomical time series datasets built on top of *PyMC*; and [tinygp](#), a JAX-based Gaussian Process modeling framework with novel algorithms for scalable linear algebra.

Co-developed and lead the maintenance of [showyourwork](#), a workflow management system for reproducible scientific publishing. Made contributions to large open source projects including *AstroPy*, *PyMC*, *JAX*, and *NumPyro*.

### The Journal of Open Source Software

Nov 2022 – present *Associate Editor-in-Chief, Astrophysics & Space Sciences*

Nov 2019 – Nov 2022 *Topic Editor, Astrophysics*

Manage a team of 10 editors to handle all submissions within the areas of Astrophysics & Space Sciences. Oversee the publication of accepted submissions, and make high-level editorial decisions for the journal.

### Department of Astronomy, University of Washington, Seattle, WA

Sep 2015 – Sep 2017 *NASA Sagan Postdoctoral Fellow*

Postdoctoral research supported by prestigious and competitive national fellowship. Developed, released, and maintained [celerite](#), an exact structure-exploiting factorization and inversion algorithm with linear scaling for a class of matrices commonly encountered in Gaussian Process modeling, and a C++/Python library for its use. Co-advised 3 PhD student theses.

### Department of Physics, New York University, New York, NY

Sep 2010 – Sep 2015 *Graduate Research Assistant*

Developed, released, and maintained several high-impact open source projects including [emcee](#), the most widely used probabilistic programming tool in astrophysics with over 10,000 citations; [george](#), a C++-accelerated Python library for scalable Gaussian Processes regression using approximate linear algebra; [corner.py](#), a popular plotting library for assessing probabilistic models with nearly 2000 citations; and [daft](#), a *matplotlib* extension for generating publication-ready probabilistic graphical models. Led the discovery of more than 20 new planets orbiting stars outside our Solar System.

## Education

---

PhD 2015, Department of Physics, New York University. Advisor: Hogg

MSc 2010, Department of Physics, Queen’s University, Canada. Advisor: Widrow

BSc 2008, Department of Physics, McGill University, Canada.

---

<sup>1</sup>Publication data collected from [NASA Astrophysics Data System](#)